

Route G2-M: A Frozen Readout Contract

Momentum first; a noisy pump tag only as a conditional alternative

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Abstract

The selected design uses the existing driven model with incoming width $\sigma_i = .1$ and momentum resolution $\sigma_d = .05$, without requiring a sideband tag. We freeze its decision rule before testing calibration errors. A separate noisy-tag channel quantifies what additional information would be required to retain coarser momentum resolution. This is a readout requirement in model units, not a built apparatus.

1 The physical choice and its boundary

All dynamics, free masses, $g = .05$, $\epsilon = .5$, $\Omega = .2$, finite internal preparation and the classical truncated-normal beam are inherited unchanged from G2-S/D. Select the same counted collisions with $m = 0$, $r = |j| = 1$. Write $s = \text{sign } j$ and $l_+ = 0$, $l_- = -2$. Their unnormalized component densities are

$$h_{sq}(x) = w_{l_s} \int_0^\infty dp f_1(p) K_{l_s,0,q}(p) \mathcal{N}(x; p_{f,sq}(p), \sigma_d^2), \quad h_s(x) = \sum_q h_{sq}(x). \quad (1)$$

Let $Z = \sum_s \int h_s dx$, $\pi_s = Z^{-1} \int h_s dx$. The same priors must be used throughout the calibration stress test: instrument response changes the readout, not the preselected event rate. Perfect mass labels, common COM frame, equal acceptance and no background contamination remain explicit assumptions.

The main choice is *momentum-only readout*, nominal $\sigma_d = .05$. It needs fewer new physical assumptions than an independent event-by-event measurement of pump work. Its nominal rule is

$$\delta_0(x) = \begin{cases} +1, & x < c, \\ -1, & x \geq c, \end{cases} \quad c = 1.25892909961. \quad (2)$$

The implemented mixture has one numerically resolved Bayes boundary. This is not a global theorem about arbitrary mixture cards. At the nominal card $P_{\text{wrong}} = .02202246905$; the majority-only error is .22530448517. The chosen overall error limit .05 is an engineering convention, not a statistical discovery threshold. The nominal false-minus rate given $j = +1$ is 1.0315%; the false-plus rate given $j = -1$ is 6.2279%. Passing the overall criterion does not mean that both signs have at least 95% recall.

The field X , previously called a detector, is a *recoil partner*; its action does not yet specify an instrument that reads its state. Likewise the G1 external sources describe linear response, not a calibrated tracking or calorimeter system. A physical energy unit, apparatus couplings, mass identification, acceptance and calibration data are needed before claiming hardware feasibility. In particular, the external model $U(1)$ has not been identified with electromagnetism. The dependence of real momentum measurements on charge, magnetic field, geometry and tracking errors is illustrated by [1]; those apparatus inputs are not implied by this prototype.

2 Unknown calibration: keep the classifier frozen

Model the true measured estimator as

$$x = (1 + a)u + b + \xi, \quad u = p_{f,sq}(p), \quad \xi \sim \mathcal{N}(0, \sigma^2). \quad (3)$$

Here a is a fractional scale error, b an offset in model momentum units, and σ the actual noise width. Declare the stress domain

$$|a| \leq .01, \quad |b| \leq .02, \quad .04 \leq \sigma \leq .06. \quad (4)$$

These are test tolerances, not measured uncertainties. For the frozen rule (2), the conditional error is exactly

$$R_0(a, b, \sigma) = \frac{1}{Z} \sum_q \int_0^\infty dp f_{.1}(p) \left[w_0 K_{0,0,q}(p) \Phi_N \left(\frac{(1+a)u_{+q} + b - c}{\sigma} \right) + w_{-2} K_{-2,0,q}(p) \Phi_N \left(\frac{c - (1+a)u_{-q} - b}{\sigma} \right) \right]. \quad (5)$$

The two terms separately give the joint probabilities for false minus and false plus. Dividing by π_+ or π_- gives the respective conditional error rates; posterior purity divides by the predicted class's total probability instead. Low aggregate error is not identical to equal performance on both signs.

If a, b were known, the invertible transformation $y = (x - b)/(1 + a)$ would reduce the oracle problem to a Gaussian noise width $\sigma/(1 + a)$:

$$R_{\text{oracle}}(a, b, \sigma) = R_{\text{Bayes}} \left(\frac{\sigma}{1 + a} \right) \leq R_0(a, b, \sigma). \quad (6)$$

Thus the oracle is independent of known b . Reoptimizing the decision rule separately at each unknown calibration value would hide the actual robustness problem, so it is reported only as a lower bound.

For a rectangular sub-box $[a_-, a_+] \times [b_-, b_+] \times [\sigma_-, \sigma_+]$, set

$$z_+(u) = (1 + a_+)u + b_+ - c, \quad z_-(u) = c - (1 + a_-)u - b_-. \quad (7)$$

Because $u \geq 0$ and Φ_N is increasing, each error term is bounded by

$$B(z) = \begin{cases} \Phi_N(z/\sigma_-), & z \geq 0, \\ \Phi_N(z/\sigma_+), & z < 0. \end{cases} \quad (8)$$

Replacing the two CDFs in (5) by these bounds gives a conservative upper bound on the risk throughout that sub-box. Different components are allowed different worst nuisances, so it may be loose. A finite cover $\{C_k\}$ yields $\sup_C R_0 \leq \max_k U(C_k)$, not a sum of the cell bounds. The numerical scan maximum is only a lower bound on this supremum. CDF integrals are quadrature-checked; analytic inequalities do not turn floating-point integration into an interval-arithmetic proof.

For this card the worst of the 27 declared grid points gives about 3.8648% at $(a, b, \sigma) = (-.01, -.02, .06)$. The unsplit bound is 5.3902%: this loose bound alone cannot certify the 5% target, but exceeding the target is not an actual failure. One predetermined $2 \times 2 \times 2$ cover of (4) tightens the envelope to about 4.46%, sufficient without repeated parameter fitting or finer scans. The implementation splits the incident quadrature at CDF-envelope kinks and checks the integral separately.

3 Alternative: a noisy independent sideband tag

Retain $\sigma_i = .1$ and compare the coarser $\sigma_d = .1$ readout with a hypothetical extra label $z \in \{-1, 0, 1\}$. Declare

$$T_\eta(z | q) = (1 - \eta)\delta_{zq} + \frac{\eta}{3}, \quad 0 \leq \eta \leq 1, \quad P(z \neq q) = \frac{2\eta}{3}. \quad (9)$$

The parameter η is mixing strength, not the mistag probability. This assumes the tag is independent of momentum noise and incoming spread conditional on true q . It is a candidate response contract, not a measurement derived from the prescribed classical pump. The joint likelihood and optimal error are

$$H_s(x, z) = \sum_q T_\eta(z | q) h_{sq}(x), \quad e_\eta = \frac{1}{Z} \sum_z \int dx \min(H_+(x, z), H_-(x, z)). \quad (10)$$

For $\eta_2 \geq \eta_1$, $\eta_1 < 1$, define $\lambda = (\eta_2 - \eta_1)/(1 - \eta_1)$. The uniform matrix $U_{zq} = 1/3$ obeys $U^2 = U$, giving

$$T_{\eta_2} = T_\lambda T_{\eta_1}. \quad (11)$$

Additional garbling cannot improve optimal inference: $\sum_z |\sum_{z'} T_\lambda(z | z') d_{z'}| \leq \sum_{z'} |d_{z'}|$. Apply this pointwise to $d_z = H_+(x, z) - H_-(x, z)$ to prove

$$e_{\text{ideal}} = e_0 \leq e_{\eta_1} \leq e_{\eta_2} \leq e_1 = e_{\text{no tag}}. \quad (12)$$

The code verifies both limiting cases and intermediate noise levels. At $\sigma_d = .1$, the passing-set endpoint lies in $\eta \in [.1758, .1768]$, corresponding to wrong-tag probability 11.72–11.78% *within this specific response family*. The endpoints bracket a numerical crossing, not calibration uncertainty. At the primary $\sigma_d = .05$, no tag is required for the overall 5% criterion.

Why aggregate tag accuracy is insufficient. In this selected ensemble $P(q = 0) = .88983936$. A useless tag that always emits $z = 0$ therefore has 88.98% raw accuracy, yet leaves the coarser-momentum error at 5.94485%. This exceeds the roughly 88.2% correct-label fraction at the symmetric-channel threshold. There is no contradiction: the constant tag has a different confusion matrix and no information. The requirement is the full conditional response $T(z | q)$, not a single accuracy percentage. These tag comparisons assume a known calibrated response and its optimal classifier; they are not a frozen-rule robustness guarantee.

Do not count the same information twice

Suppose the proposed tag is merely a deterministic or noisy processing of the same measured x : $H_s(x, z) = h_s(x)T(z | x)$. Then, identically,

$$\sum_z \min(h_+(x)T(z | x), h_-(x)T(z | x)) = \min(h_+(x), h_-(x)). \quad (13)$$

The Bayes error cannot improve. Computing a pump label from the same outgoing momentum and calling it an independent tag is therefore invalid. A drive clock records phase, not the event's absorbed energy. Even a genuinely additional sensor needs its joint response and correlations modeled; it does not automatically satisfy T_η . For two distinct sensors sharing the same true momentum, the correct likelihood integrates that latent momentum once: $h_s(x, y) = \sum_q w_{ls} \int f K_{ls,0,q} R_1(x | u) R_2(y | u) dp$. Multiplying separately marginalized $h_s(x)h_s(y)$ is generally wrong.

4 Closure and experimental handoff

The accompanying reproducible reports give the nominal confusion matrix, frozen calibration risks, conservative box bounds and noisy-tag requirements. No action parameters are retuned. The primary choice remains the momentum-only contract; the noisy tag is an alternative requirement if finer momentum resolution is unavailable.

The bounded mathematical step ends here. To proceed physically, choose the energy scale and readout interactions or supply an actual system's response/beam calibration. Do not replace this missing physical choice with indefinitely finer Gaussian scans. Absolute collision probability is still unnecessary for this conditional discrimination question. Only a request for absolute yields, finite-pulse effects or coherent interference calls for four-dimensional packets, timing and a pump envelope.

References

- [1] Particle Data Group, *Particle Detectors at Accelerators*, Review of Particle Physics (2024), section 35.13, Official review (PDF).